Telecom Customer Churn Prediction and Retention Intelligence Using Machine Learning

Project Report
Course: AI for Engineers

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DECLARATION

We hereby declare that the project report entitled “Telecom Customer Churn Prediction and Retention Intelligence Using Machine Learning” is an authentic record of our own work, carried out as part of the course AI for Engineers at Thapar Institute of Engineering and Technology, Patiala, under the guidance of Arun Pundir. The synthetic dataset generator, the published methods and the software libraries used in this report have been acknowledged in the references.

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ABSTRACT

Customer churn, the loss of subscribers who stop using a service, is one of the largest sources of lost revenue for telecom operators, because a customer who leaves must be replaced by a new one, and a new customer is usually harder to win than an existing one is to keep. If the operator knows in advance which customers are likely to leave, it can contact them with a suitable offer before they go. This project studies whether the churn of a telecom customer can be predicted from information that the operator already holds, and how such a prediction can be turned into practical retention decisions.

We generate a synthetic dataset of 20,000 telecom customers with 13 attributes (tenure, monthly charges, contract type, internet service, payment method, senior-citizen flag, number of services, support calls, complaints, late payments, data usage, call minutes and satisfaction score). The target is a binary churn label; 35.9% of the customers churn. The data is divided into a training set (16,000 customers) and a test set (4,000 customers) with a stratified split, so that both parts have the same churn rate. Three models are compared on the same split: Logistic Regression, Random Forest and XGBoost.

On the unseen test customers, Logistic Regression achieved the best ROC-AUC of 0.8079 (accuracy 72.3%, recall 72.6%, precision 59.4%, F1 0.653), followed closely by XGBoost (0.8032) and Random Forest (0.7981). The 95% bootstrap intervals of the three models overlap, so the report does not claim that one model is clearly superior. SHAP values show that contract type, tenure, satisfaction, support calls and complaints are the strongest drivers. The predicted probability is converted into three risk levels (Low, Medium, High), a customer health score and rule-based retention actions; the High-risk group (one third of the test customers) contains 60.5% of all churners, and its actual churn rate is 64.5% against 9.5% in the Low-risk group.

The data in this report is synthetic, so the numbers show that the pipeline works and how it should be evaluated; they are not findings about any real telecom company. The report explains the business background, the dataset, the method, the evaluation metrics and the real-world importance of the problem, analyses the results in detail, and describes a Streamlit dashboard prototype.

Keywords: customer churn, telecom, classification, Logistic Regression, Random Forest, XGBoost, SHAP, explainability, retention, class imbalance.

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CHAPTER 1
INTRODUCTION

1.1 Background

A telecom operator earns its revenue from subscribers who pay every month for calls, data and other services. When a subscriber cancels the service or moves to a competitor, the operator loses that income. This event is called customer churn, and the churn rate is the share of customers who leave in a given period. Because most telecom markets are saturated, the number of new customers who can be won is limited, and keeping the existing customers has become as important as acquiring new ones.

Operators already collect a large amount of information about each customer: the type of contract, how long the customer has been with the company, what the customer pays, how often the customer calls the support centre, and so on. Machine learning can learn from the customers who left in the past and estimate, for every present customer, the probability of leaving. The operator can then spend its limited retention budget on the customers who are most at risk, instead of giving the same offer to everyone.

1.2 What is customer churn prediction?

Churn prediction is a supervised binary classification problem. Each customer is described by a row of features, and the label is 1 if the customer churned and 0 if the customer stayed. A model that outputs a probability is more useful than one that outputs only a yes or no, because the probability can be ranked, grouped into risk levels and combined with business rules. This project therefore treats the predicted probability as the main output and builds three further layers on it: a risk level, a customer health score and a recommended retention action.

1.3 Problem statement

To develop a machine-learning system that predicts whether a telecom customer will churn from 13 attributes held by the operator, to explain which attributes drive the prediction, and to convert the prediction into risk levels, a health score and retention actions, with accuracy evaluated on customers that were not used for training.

1.4 Objectives

- To generate and understand a telecom customer dataset and prepare it for machine learning.
- To study the relationship between customer attributes and churn using exploratory data analysis.
- To select input features carefully and avoid target leakage.
- To split the data in a stratified way so that the churn rate is the same in the training and the test set.
- To build and compare Logistic Regression, Random Forest and XGBoost.
- To evaluate the models using accuracy, precision, recall, F1, ROC-AUC, PR-AUC and the Brier score, and to analyse what the errors mean for the business.
- To explain the predictions with SHAP and to build risk levels, a health score and retention rules on top of them.

- To prototype a Streamlit dashboard that uses the saved model.

1.5 Scope

The project is an offline study on a synthetic dataset. It estimates the probability of churn from the selected inputs; it does not by itself prove that a retention action will keep a customer, and the retention rules have not been tested on real customers. The dashboard described in Chapter 8 is a prototype only.

1.6 Relation to the course

AI for Engineers deals with applying AI methods to engineering systems. A telecom network and its customer base form such a system: data is produced by billing and support systems, domain knowledge is used to choose inputs, and the model is judged with engineering questions in mind, such as how many customers must be contacted to reach most of the churners, and how many of the contacts will be unnecessary.

1.7 Organisation of the report

Chapter 2 reviews related work and the real-world importance of the problem. Chapter 3 describes the dataset and Chapter 4 the business background. Chapter 5 explains the method, Chapter 6 the models and metrics, and Chapter 7 presents and analyses the results. Chapter 8 describes the prediction application, Chapter 9 the limitations and future work, and Chapter 10 concludes the report.

CHAPTER 2
LITERATURE REVIEW AND REAL-WORLD RELEVANCE

2.1 Importance of customer churn

The effect of churn on a subscription business can be seen from a simple relation. If a constant share c of the customers leaves every month, the expected time that a customer stays is about $1/c$ months. A monthly churn of 5% therefore corresponds to an expected lifetime of 20 months, while a monthly churn of 10% halves it to 10 months. This is shown in Figure 1.

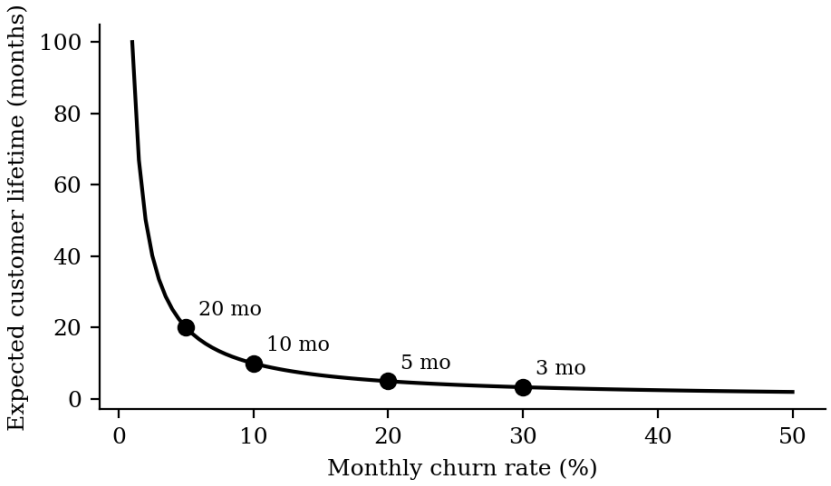


Figure 1: Expected customer lifetime against monthly churn rate (lifetime = 1 / churn rate).

Figure 1 is calculated from the formula lifetime = 1 / churn rate and assumes that the churn rate is constant over time, which is only an approximation; real churn depends on tenure and contract. It shows why even a small reduction in churn is valuable: every month of extra lifetime is another month of revenue from the same customer. It is also commonly stated in the industry that winning a new customer costs more than keeping an existing one; the ratio differs from company to company, so no single figure is used in this report.

2.2 Applications

Table 1: Applications of churn prediction.

Application	**Why churn matters**	**Benefit of prediction**
Mobile and broadband operators	Customers can switch provider easily at the end of a contract	Early offer to customers at risk, before they leave
Subscription media and streaming	Monthly cancellation is cheap and frequent	Targeted content or price offers to unhappy users
Banks and insurance	Lost accounts reduce long-term income	Relationship calls to customers whose activity falls
Customer support teams	Repeated complaints often come before cancellation	Priority handling of high-risk customers
Marketing planning	Retention budget is limited	Spend the budget on customers where it matters most

2.3 Related work and methods used

Telecom churn is a well-known benchmark problem, and public sample datasets such as the IBM telco customer-churn sample (about seven thousand customers) are widely used for teaching and for comparing methods. Typical studies compare simple linear models with tree-based models on tabular customer data, and report accuracy, recall and ROC-AUC.

Three methods are used in this project. Logistic Regression is the classical baseline and is easy to interpret. Random Forest (Breiman, 2001) averages many decision trees trained on random parts of the data and can capture non-linear effects and interactions. XGBoost (Chen and Guestrin, 2016) builds trees one after another so that each tree corrects the errors of the earlier ones, and it is often a strong model on tabular data. For explanation, SHAP (Lundberg and Lee, 2017) assigns to every feature a contribution to each individual prediction. Class imbalance is a common issue in churn studies; techniques such as oversampling (SMOTE, Chawla et al., 2002) and class weights are used to handle it, and this project uses class weights.

2.4 Observations from the literature

- Different studies use different datasets, features and metrics, so numbers from different papers cannot be compared directly with ours.
- Accuracy alone is misleading for churn, because the churners are the minority; precision, recall and ROC-AUC are needed as well.
- On tabular data with mostly additive effects, a well-tuned linear model can be as good as a tree ensemble; the extra complexity should be justified by a measured gain.
- A prediction is only useful if it can be explained and linked to an action, which is why this project adds SHAP, risk levels and retention rules.

CHAPTER 3
DATASET DESCRIPTION

3.1 Source

A real operator's customer data is confidential, and the public samples are small. For this reason the dataset used in this project is generated by a Python script (random seed 42) that simulates 20,000 telecom customers. The attributes are drawn from distributions chosen to look realistic, and the churn label is then drawn from a probability that depends on the attributes (longer tenure, longer contracts and higher satisfaction lower the probability; month-to-month contracts, fibre service, support calls, complaints, late payments and electronic-cheque payment raise it) with random noise added, so that the problem is not trivially separable.

Table 2: Main facts about the dataset.

Item	**Value**
Number of customers (rows)	20,000
Number of raw features	13
Target	churn (1 = customer left, 0 = stayed)
Churn rate	35.9% (7,179 churners)
Columns after one-hot encoding	20 model inputs (3 categorical features become 10 columns)
Missing values introduced	400 cells (1% of monthly_charges and of data_usage_gb)
Duplicate rows	0
Generation	Python script, random seed 42

The exact numbers of any real operator will be different, and the dataset should be replaced by real data before any business use. The generator script and the saved CSV file are kept with the project so that the experiment can be repeated.

3.2 Columns of the dataset

Table 3: Columns and their role in this project.

Column	Meaning	Role
tenure	Months the customer has been with the operator	Input
monthly_charges	Monthly bill amount	Input
contract	Month-to-month, One year or Two year	Input (categorical)
internet_service	Fiber, DSL or None	Input (categorical)
payment_method	Electronic check, Credit card, Bank transfer or Mailed check	Input (categorical)
senior_citizen	1 if the customer is a senior citizen	Input
num_services	Number of services subscribed	Input
support_calls	Calls to the customer-care centre in the recent period	Input
complaints	Formal complaints in the recent period	Input
late_payments	Number of late payments	Input
data_usage_gb	Monthly data usage in GB	Input
call_minutes	Monthly call minutes	Input
satisfaction	Satisfaction score from 1 (low) to 5 (high)	Input
churn	Whether the customer left	Target

3.3 One row per customer

Every row describes one customer, and customers are independent of each other in this dataset. This is different from time-series data, where neighbouring rows are almost identical and the split must be done by session. Here a random split at customer level is appropriate, provided that the churn rate is kept the same in both parts (Section 5.5).

3.4 Size of the data

With 20,000 customers and 20 model input columns, the data has about 400 thousand values and is small enough to train all three models in seconds on a laptop. About 7,179 customers churn, which is enough for the models to learn from, but the churners are still the minority class.

3.5 Points to remember

- The data is synthetic, so the relations found in it are the relations that were built into the generator plus noise; they are not facts about any real operator.
- The attributes are a snapshot of each customer; there is no month-by-month history.
- The satisfaction score and the support-call count are assumed to be known before the customer decides to leave; if they were recorded after the decision, they would leak the target (Section 5.4).

CHAPTER 4
BUSINESS AND DOMAIN BACKGROUND

4.1 Why customers leave

Customers usually leave a telecom operator for a small number of reasons: the price is higher than at a competitor, the service quality is poor, a problem was not solved by customer care, or the contract has ended and nothing keeps the customer. Contract type is particularly important. A customer on a month-to-month contract can leave at any time, while a customer on a two-year contract is bound, and often receives a discount, so the churn rate of month-to-month customers is expected to be much higher.

4.2 Groups of features

The 13 features of this project follow the picture in Figure 2. Account and contract features (contract, tenure, monthly charges, number of services, internet service) describe the commercial relationship. Behaviour features (data usage, call minutes, late payments, payment method) describe how the customer uses and pays for the service. Service-experience features (support calls, complaints and satisfaction) describe how the customer feels about the operator.

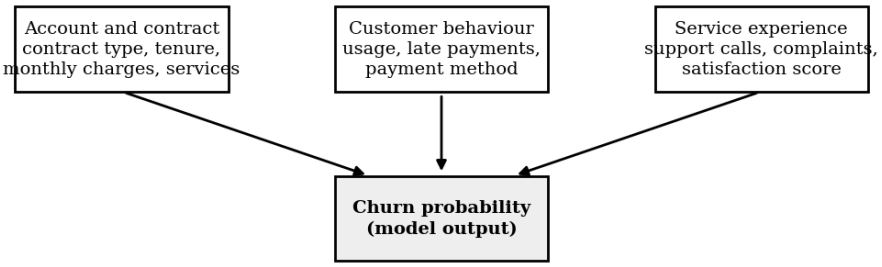


Figure 2: Main factors that determine the churn probability of a customer.

4.3 Class imbalance and the cost of errors

In the dataset 35.9% of the customers churn, so a model that always predicts “stays” would be right for 64.1% of the customers and would be completely useless, because it never finds a churner. For this reason accuracy cannot be the only metric. The two kinds of error have different costs. A false negative (a churner predicted to stay) means that the customer leaves without any attempt to keep him. A false positive (a loyal customer predicted to churn) means that a retention offer is given to someone who did not need it, which costs money but does not lose the customer. Which error is more serious depends on the cost of the offer and on the value of the customer, and the decision threshold can be changed to match it (Section 7.4).

CHAPTER 5
METHODOLOGY

The overall workflow of the project is shown in Figure 3.

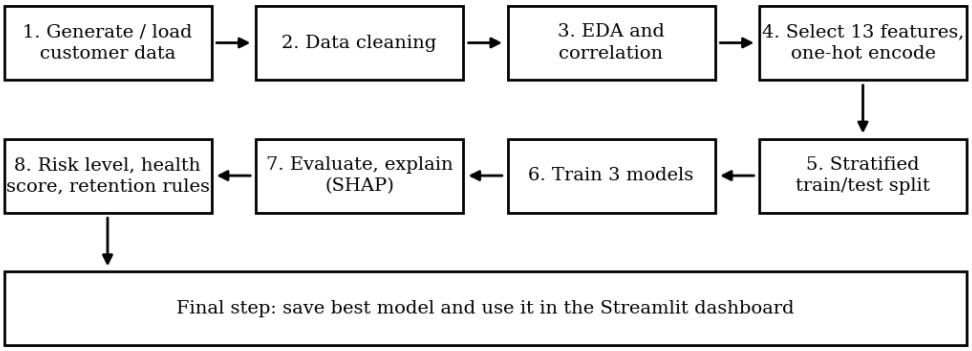


Figure 3: Workflow of the project.

5.1 Data cleaning

The data is first checked for shape, data types, missing values and duplicate rows, and basic statistics (minimum, maximum, mean and standard deviation) are calculated. The check found 400 missing values, which are 1% of the cells in monthly_charges and in data_usage_gb, and 0 duplicate rows. The missing values are filled with the median of the training set; the same training median is used for the test set so that no information from the test customers enters the training.

5.2 Exploratory data analysis

The analysis is question-driven. We look at the balance of the two classes, the churn rate in each contract type, the distribution of tenure for churners and non-churners, and the relation of churn to support calls, complaints and satisfaction. Figure 4 shows the class balance and the churn rate by contract, and Figure 5 shows the distribution of tenure.

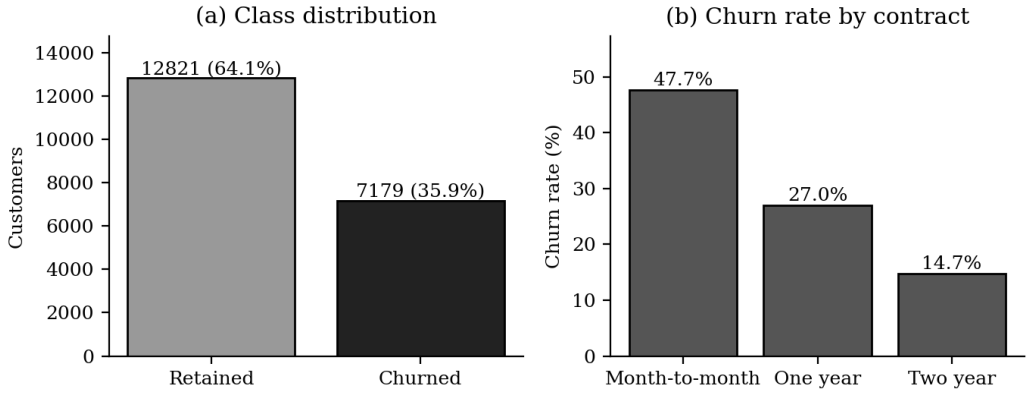


Figure 4: Class distribution and churn rate by contract type.

Churn is strongly linked to the contract: 47.7% of the month-to-month customers churn, against 27.0% of the one-year customers and 14.7% of the two-year customers.

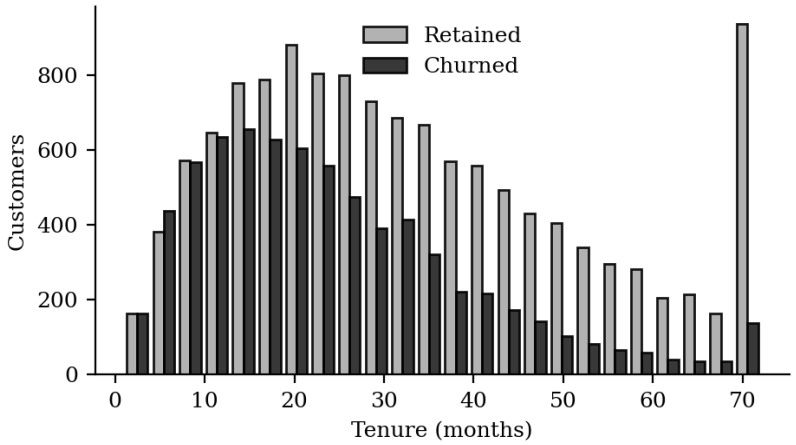


Figure 5: Distribution of tenure for retained and churned customers.

The churners are concentrated at short tenure, which agrees with the general experience that the first months are the most risky period of a customer relationship.

5.3 Correlation analysis

Correlation measures the strength of a straight-line relationship between two variables on a scale from −1 to +1. For a binary target it shows whether a higher value of a feature goes with a higher or a lower chance of churn. The values obtained for the churn label are given in Table 4 and Figure 6.

Table 4: Correlation of the strongest variables with churn.

Variable	**Correlation**	**Comment**
satisfaction	−0.28	Happier customers stay
contract_Month-to-month	+0.27	Free to leave at any time
support_calls	+0.26	Repeated calls signal a problem
tenure	−0.24	Long-standing customers stay
contract_Two year	−0.22	Bound by a long contract
complaints	+0.21	Unresolved problems
internet_service_Fiber	+0.17	Higher bills and expectations
contract_One year	−0.11	Some protection
internet_service_DSL	−0.10	Lower churn than fibre
internet_service_None	−0.09	Weak
monthly_charges	+0.09	Higher bill, higher risk
late_payments	+0.09	Payment difficulty
payment_method_Electronic check	+0.08	Higher churn than other methods

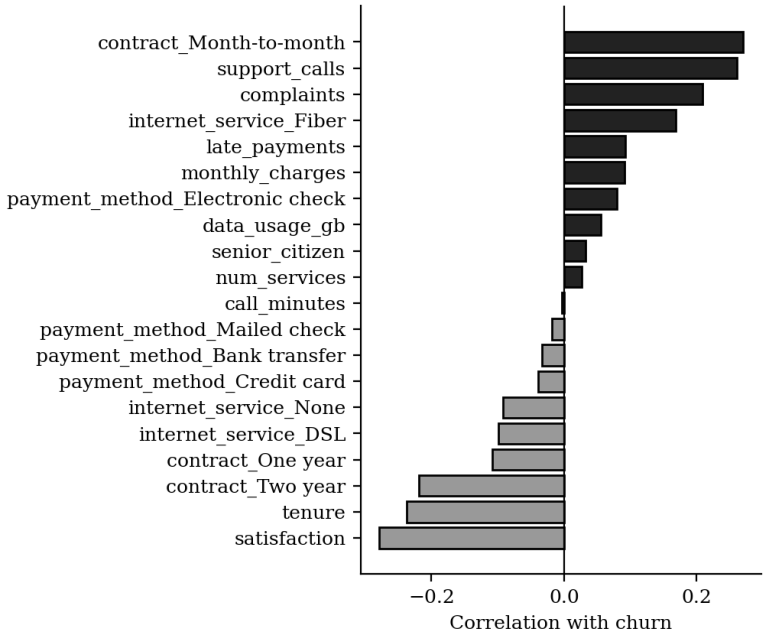


Figure 6: Correlation of each model input column with churn.

The table shows an important point for the later analysis: no single feature has a correlation larger than about 0.3 with churn. Churn is therefore the result of many small influences acting together, not of one dominant cause, and a model has to combine several features to predict it. Correlation only measures straight-line association and does not prove cause and effect; a feature with a low correlation, such as num_services, can still be useful when combined with others.

5.4 Feature selection and target leakage

All 13 attributes are used as inputs and churn is the target. Target leakage means that a feature contains information that would not be available at the time of prediction or that is a consequence of the target itself, for example a “reason for leaving” field or the date of cancellation. Such columns are not present in this dataset and must be excluded if real data is used. The satisfaction score, the support-call count and the complaint count are used on the assumption that they are recorded before the customer decides to leave.

5.5 Stratified train/test split

The customers are divided into a training set of 16,000 (80%) and a test set of 4,000 (20%) with train_test_split(stratify=y, random_state=42). Stratification keeps the churn rate the same in both parts (35.9% in the full data and 35.9% in the test set), which matters because the churners are the minority class and a random draw could give a test set with a different class balance (Figure 7).

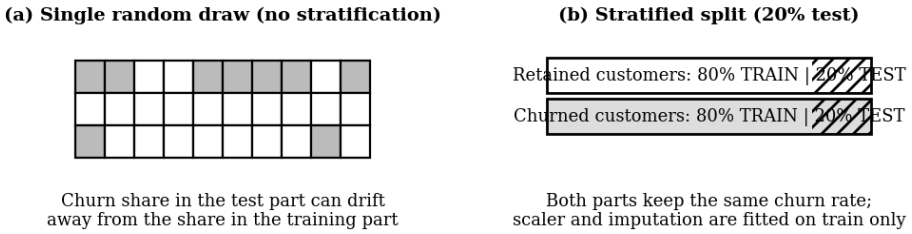


Figure 7: Plain random split compared with a stratified split.

The test set represents customers that the model has never seen. Missing values, the scaler used by Logistic Regression and the one-hot encoding columns are fitted on the training data only, and the test set is used only for the final evaluation.

5.6 Reproducibility

The random seed (42), the test fraction, the model settings and the list of input columns are fixed, so that the experiment can be repeated and gives the same numbers.

CHAPTER 6
MACHINE-LEARNING MODELS AND EVALUATION METRICS

6.1 Learning task

The task is supervised binary classification. The model receives the feature vector X of a customer and learns to predict y, the churn label, and the probability P(churn = 1 | X).

X = [tenure, monthly_charges, contract, internet_service, payment_method, senior_citizen, num_services, support_calls, complaints, late_payments, data_usage_gb, call_minutes, satisfaction], y = churn

6.2 Models

Table 5: Models used in the project.

Model	Idea	Reason for use
Logistic Regression	Weighted sum of the inputs passed through a sigmoid: $P = 1 / (1 + e^{-(b0 + b1x1 + \ldots + bkxk)})$	Simple, interpretable baseline
Random Forest	Average of many decision trees trained on random parts of the data (300 trees)	Captures non-linear effects and interactions
XGBoost	Trees built one after another, each correcting earlier errors (300 trees, depth 4, learning rate 0.05)	Strong tabular model, works with SHAP

All models are trained on the same training customers and tested on the same test customers with the same metrics. Class imbalance is handled with class weights (balanced weights for Logistic Regression and Random Forest, and scale_pos_weight for XGBoost), and the decision threshold is 0.5 unless stated otherwise. A model is called better only after all models have been measured under identical conditions.

6.3 Logistic Regression

Logistic Regression fits one coefficient for each input so that the log-loss on the training data is minimal; the inputs are standardised first so that the coefficients can be compared. It can represent only an additive effect of the inputs on the log-odds of churn. This is a limitation if the true relation contains strong interactions, but the model is valuable as a baseline: any more complex model must clearly improve on it to justify its extra complexity.

6.4 Evaluation metrics

The metrics are defined with the confusion matrix, whose four counts are true positives (TP, churners found), false positives (FP, loyal customers flagged), false negatives (FN, churners missed) and true negatives (TN).

- **Accuracy** = (TP + TN) / (TP + TN + FP + FN): the share of correct predictions.
- **Precision** = TP / (TP + FP): of the customers flagged as churners, the share who really churn.
- **Recall** = TP / (TP + FN): of the real churners, the share that is found.
- **F1** = 2 · Precision · Recall / (Precision + Recall): the balance of the two.
- **ROC-AUC**: the probability that a randomly chosen churner receives a higher score than a randomly chosen non-churner. 0.5 means random ranking and 1 means perfect ranking.
- **PR-AUC** (average precision): the area under the precision-recall curve; it is more sensitive to the minority class than ROC-AUC.
- **Brier score** = mean of $(p - y)^2$: the quality of the predicted probabilities; lower is better.

Because the evaluation uses 4,000 test customers, the metrics have some random uncertainty. To show it, the ROC-AUC is also given with a 95% interval obtained by bootstrap resampling of the test set (300 resamples).

6.5 Explainability with SHAP

SHAP (SHapley Additive exPlanations) divides the difference between a customer's prediction and the average prediction among the features. The mean absolute SHAP value of a feature over many customers shows how strongly the model relies on it, and the sign of the value for an individual customer shows whether the feature pushes that customer towards churn or towards staying. As with correlation, the explanation shows what the model uses and does not prove that the feature causes churn.

6.6 Risk levels, health score and retention rules

The probability of the chosen model is converted into three decision layers. **Risk level:** Low if the probability is below 0.30, Medium from 0.30 to 0.60, and High at 0.60 or more. **Health score:** 100 × (1 − probability), a number from 0 to 100 in which a high value means a healthy customer. **Retention action:** a small set of rules that combines the risk level with the customer's own attributes, given in Table 6.

Table 6: Rule-based retention actions.

Condition	Recommended action
High risk and contract is month-to-month	Offer a discounted 12-month contract
High risk and (support_calls ≥ 3 or complaints ≥ 2)	Priority call-back by a senior support agent
High risk and internet_service is Fiber and satisfaction ≤ 2	Service-quality check and technical visit
Medium risk and late_payments ≥ 2	Flexible payment date or auto-pay incentive
Medium risk and tenure < 12 months	Welcome-journey follow-up and usage tips
Low risk and tenure ≥ 36 months	Loyalty reward; no discount needed

The thresholds and rules are design choices of this project, written down so that they can be reviewed and changed by the business; they have not been tested on real customers, so no claim is made about how much churn they would prevent.

CHAPTER 7
RESULTS AND ANALYSIS

7.1 Experimental setup

Thirteen customer attributes were used as inputs and the churn label as the target. The dataset of 20,000 customers was divided into 16,000 training and 4,000 test customers with a stratified split. Models were trained on the training customers and all reported metrics are calculated on the held-out test customers at the threshold 0.5.

7.2 Model comparison

The three models were evaluated under identical conditions. Logistic Regression obtained the highest ROC-AUC (0.8079), followed by XGBoost (0.8032) and Random Forest (0.7981). The results are given in Table 7 and Figures 8 and 9.

Table 7: Measured results of the three models on unseen test customers (threshold 0.5).

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	0.7232	0.7332	0.7282
Precision	0.5937	0.6244	0.6027
Recall	0.7256	0.6448	0.7131
F1	0.6531	0.6345	0.6533
ROC-AUC	0.8079	0.7981	0.8032
PR-AUC	0.7041	0.6846	0.6971
Brier score	0.1798	0.1770	0.1804
ROC-AUC 95% interval	0.795 – 0.821	0.784 – 0.812	0.792 – 0.817

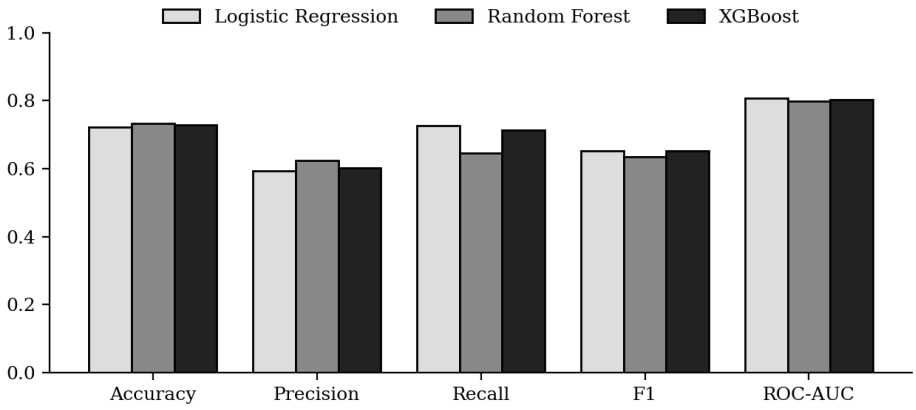


Figure 8: Comparison of the three models on the main metrics.

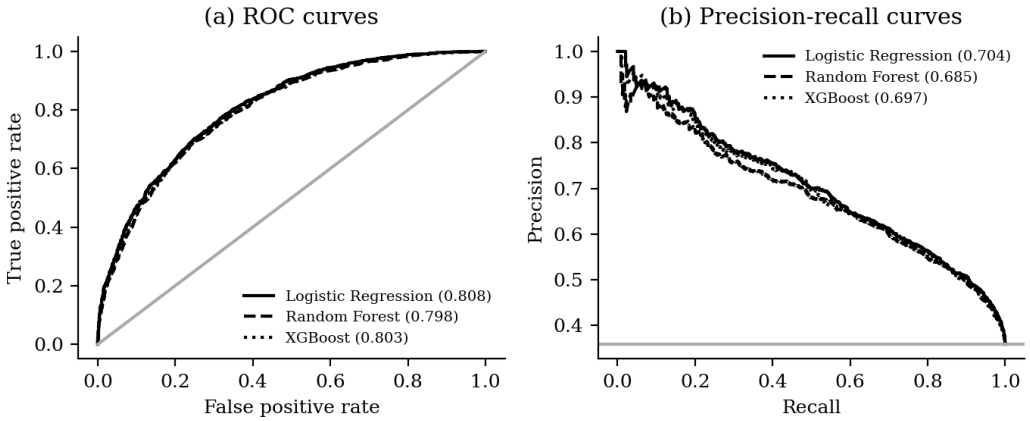


Figure 9: ROC curves and precision-recall curves of the three models (area under each curve in brackets).

7.3 Further analysis of the results

Table 8: Quantities derived from the reported results (Logistic Regression).

Quantity	**Value**	**How obtained**
Accuracy of “nobody churns”	64.1 %	1 − 35.9 % churn rate
Accuracy of the model	72.3 %	(1042 + 1851) / 4,000
Specificity (loyal customers recognised)	72.2 %	1851 / (1851 + 713)
False positive rate	27.8 %	713 / (1851 + 713)
Customers flagged as churners	1755 (43.9 %)	1042 + 713
Flagged customers who do not churn	40.6 %	1 − precision
Brier score of a constant prediction	0.230	p × (1 − p), p = churn rate
Churn rate in High-risk group	64.5 %	180 % of the average rate (lift 1.80)
Churners found in High-risk group	60.5 %	from 33.7 % of the customers
Churners found in High + Medium	91.1 %	from 66.2 % of the customers

The values in Table 8 are calculated from the confusion matrix and the risk-level counts and show more than the headline metrics.

Observation 1: the models learn a real part of the pattern

A model that predicts “stays” for everybody reaches an accuracy of 64.1%, with a recall of zero. Logistic Regression reaches 72.3% accuracy and, more importantly, finds 72.6% of the churners. Its ROC-AUC of 0.808 means that a randomly chosen churner gets a higher score than a randomly chosen loyal customer about 81% of the time, against 50% for random ranking, and its Brier score (0.180) is clearly below that of a constant prediction (0.230). The customer attributes therefore carry usable information about churn.

Observation 2: the three models are close, and no winner can be claimed

The ROC-AUC of the three models differs by only 0.010 between the best and the worst, and the 95% bootstrap intervals overlap (Table 7). The ranking Logistic Regression, XGBoost, Random Forest is therefore the ranking of the point estimates on this test set, and it could change on another sample. The non-linear models gave no measurable gain over the linear baseline. A probable reason is that in this synthetic dataset the effects were built in mostly as additive terms plus random noise, which a logistic model represents well; this is a hypothesis and not a proven result, and on real operator data tree models may do better. On the evidence of this report, Logistic Regression is a reasonable choice because it is simple, fast and interpretable.

Observation 3: the precision-recall trade-off

At the threshold 0.5 the chosen model finds 72.6% of the churners with a precision of 59.4%, which means that about 41 of every 100 flagged customers would receive an offer without needing it. Changing the threshold moves the balance, as Figure 11 shows; the F1 score is highest, at 0.661, near a threshold of 0.45, but the curve is flat, so the business cost of an offer and the value of a saved customer should decide the threshold rather than F1.

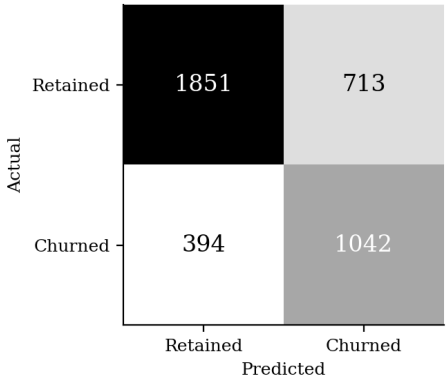


Figure 10: Confusion matrix of the Logistic Regression model on the test customers (threshold 0.5).

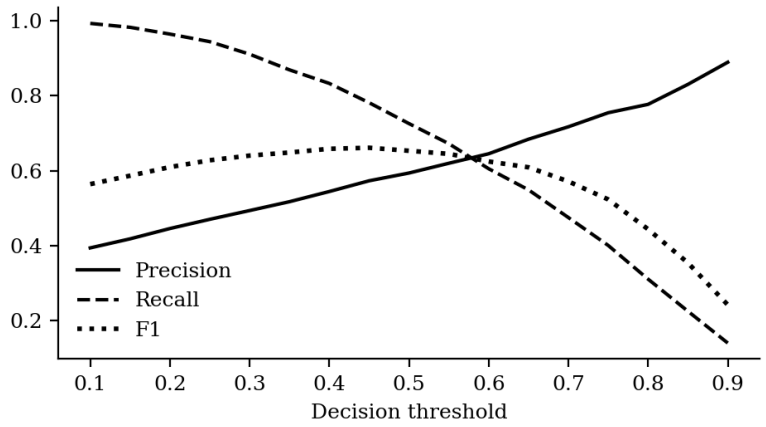


Figure 11: Precision, recall and F1 of the model at different decision thresholds.

Observation 4: the risk levels are useful for targeting

The three risk levels separate the customers well. The actual churn rate is 9.5% in the Low-risk group, 33.7% in the Medium group and 64.5% in the High group (Table 9, Figure 12), so the levels are ordered correctly and the probabilities mean what they say. If the operator contacts only the High-risk group, which is 33.7% of the customers, it reaches 60.5% of all churners; adding the Medium group reaches 91.1% of the churners but needs 66.2% of the customers to be contacted.

Table 9: Risk levels on the test customers.

Risk level	Probability range	Customers	Actual churn rate
Low	below 0.30	1350	9.5 %
Medium	0.30 to 0.60	1303	33.7 %
High	0.60 or more	1347	64.5 %

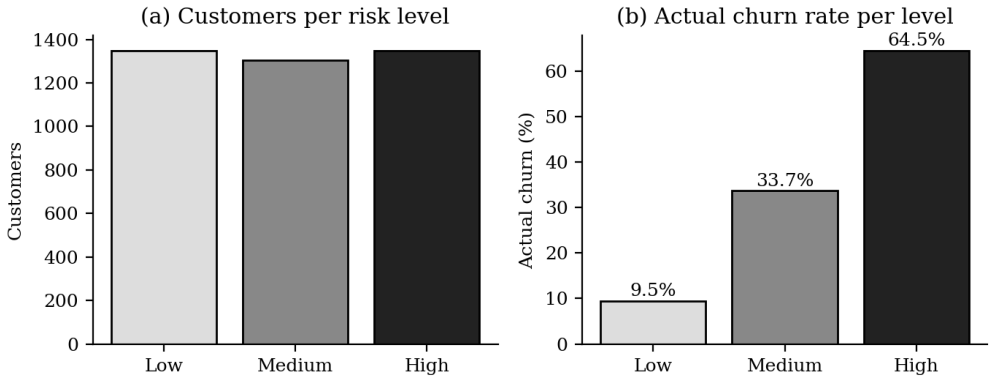


Figure 12: Number of customers and actual churn rate in each risk level.

The health score (100 × (1 − probability)) behaves as expected: its average is 64.3 for the customers who stayed and 35.3 for the customers who churned (Figure 13).

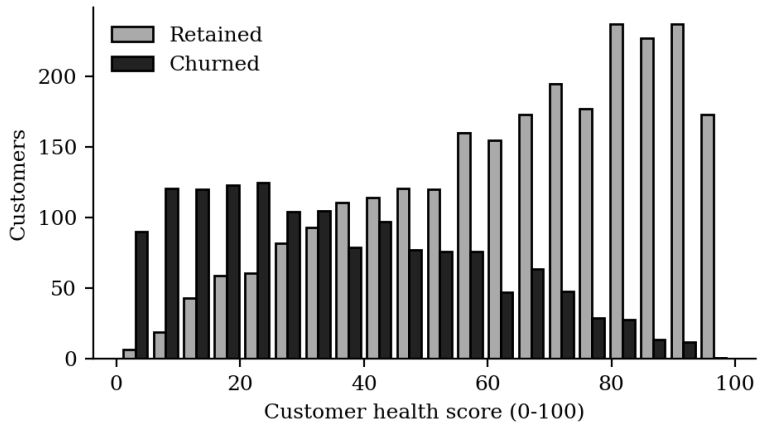


Figure 13: Distribution of the customer health score for retained and churned customers.

Observation 5: what drives churn in the model, and probable reasons for the remaining error

The SHAP analysis of the XGBoost model (Figures 14 and 15) shows that the month-to-month contract (mean |SHAP| 0.61), tenure (0.55), satisfaction (0.35), support calls (0.30) and complaints (0.22) are the strongest drivers. The standardised coefficients of the Logistic Regression model give the same picture: tenure (−0.70), the two-year contract (−0.47), the month-to-month contract (+0.47), satisfaction (−0.41) and support calls (+0.36). Two different methods agree. Because the data is synthetic, this agreement shows that the pipeline recovers the relations that were put into the generator, which is a useful check of the method, but it is not evidence about real customers.

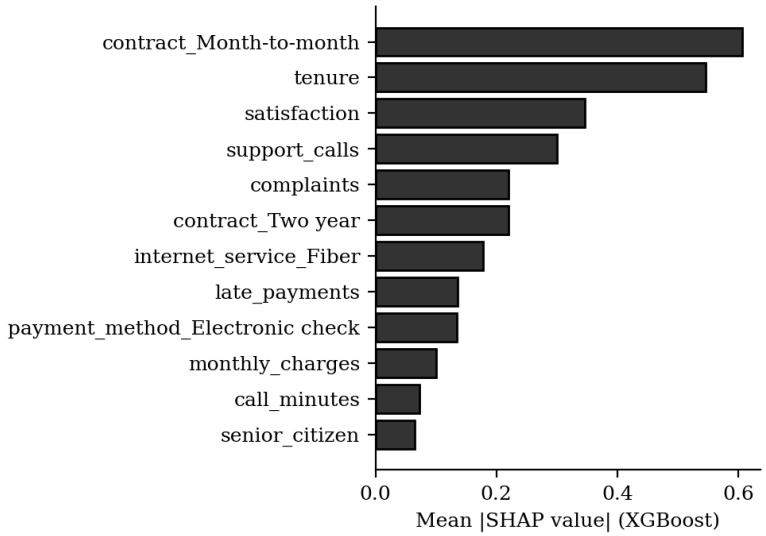


Figure 14: Mean absolute SHAP value of the most important input columns (XGBoost).

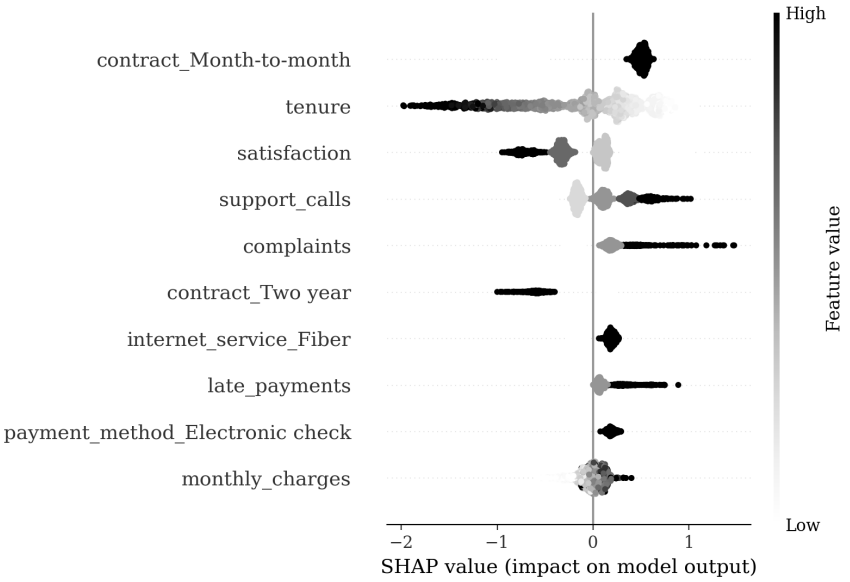


Figure 15: SHAP summary plot of the most important input columns (XGBoost).

In Figure 15 each point is a customer, and its horizontal position shows whether the feature pushes the prediction towards churn (right) or towards staying (left); the colour shows the value of the feature.

About 27.7% of the customers are still misclassified, and the ROC-AUC is 0.81 and not close to 1. We think that the following reasons contribute. These are hypotheses and are not proven by the present results.

- Random noise: the churn label includes a random component, so even a perfect model cannot reach 100% accuracy.
- No history: the models see a snapshot of the customer and not the trend of usage, bills or support calls over the last months, which in practice is a strong signal.
- Missing information: real customers also leave for reasons that are not in the data, such as a competitor's offer or a move to another city.
- Threshold and class weights: the 0.5 threshold with balanced weights favours recall over precision; a different cost balance would change the metrics.

7.4 Threshold and cost

The choice of threshold is a business decision. If an offer is cheap compared with the value of a customer, the operator can lower the threshold, find more churners and accept more false alarms. If the offer is expensive, a higher threshold gives a shorter, more accurate list. Figure 11 shows how precision and recall move in opposite directions as the threshold changes, and the risk levels of Section 6.6 are a simple way of presenting this choice to non-technical staff.

7.5 Planned further analysis

- Tune the hyper-parameters of Random Forest and XGBoost with cross-validation and report the spread of the results over several random splits.
- Calibrate the probabilities and check calibration plots.

- Estimate the profit of retention campaigns for different thresholds, using real offer costs and customer values.
- Repeat the experiment on a real or public telecom churn dataset.

CHAPTER 8
PREDICTION APPLICATION

8.1 Idea

After the model comparison is completed, the chosen model, the scaler and the list of columns are saved to a file with joblib. A small dashboard built with Streamlit loads the file, accepts the customer details from the user and displays the churn probability, the risk level, the health score, the recommended retention action and the main SHAP drivers of that customer.

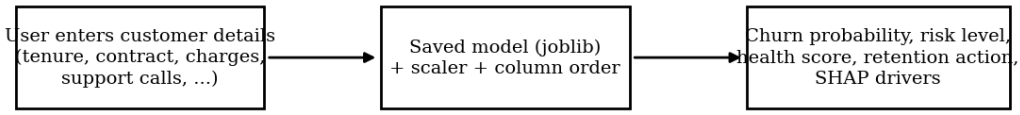


Figure 16: Architecture of the prediction application.

8.2 Working

1. The user enters the customer attributes: tenure, charges, contract, internet service, payment method, services, support calls, complaints, late payments, usage and satisfaction.
2. The values are encoded and scaled in the same way and in the same column order as in training.
3. The model returns the churn probability, which is converted into the risk level, the health score and the retention action and is shown on the screen.

8.3 Tools used

- Python, Pandas and NumPy for data handling.
- Matplotlib for graphs.
- Scikit-learn for Logistic Regression, Random Forest, splitting and metrics; XGBoost for the boosted model.
- SHAP for explanation.
- Joblib for saving the model and Streamlit for the interface.

8.4 Use with care

The application only shows an estimate. Considering that about 41 out of every 100 flagged customers do not churn, the output should be used to prioritise contacts and not to take irreversible decisions such as cancelling a service or refusing a customer. The model was trained on synthetic data and must be retrained and validated on the operator's own data before any real use, and the use of customer data must follow the applicable privacy rules.

8.5 Script for the experiment

The core of the experiment script is given below. The complete script, which also produces the graphs and the results table, is submitted with this report.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler

X = pd.get_dummies(df.drop(columns='churn'),
                   columns=['contract','internet_service','payment_method'])
y = df['churn']
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.2,
                                      stratify=y, random_state=42)
Xtr = Xtr.fillna(Xtr.median()); Xte = Xte.fillna(Xtr.median())
sc = StandardScaler().fit(Xtr)
lr = LogisticRegression(max_iter=1000, class_weight='balanced')
lr.fit(sc.transform(Xtr), ytr)
p = lr.predict_proba(sc.transform(Xte))[:, 1]
risk = pd.cut(p, [0, .3, .6, 1], labels=['Low','Medium','High'])
health = 100 * (1 - p)
```

CHAPTER 9
LIMITATIONS AND FUTURE WORK

9.1 Limitations

- Synthetic data: the dataset is generated by our own script, so the results show that the method works and not how well it would work for a real operator. The relations in the data are those that we built in.
- Snapshot data: the models predict from one row per customer and cannot use the trend of usage, bills or complaints over time.
- Accuracy: with 72.3% accuracy and 59.4% precision, about four out of ten flagged customers would not have churned.
- Single split: the results come from one train/test split; the bootstrap interval shows the test-set uncertainty but not the variation between splits.
- Untested retention rules: the retention actions are rules written by us and their effect on real churn has not been measured.
- Offline study: all results come from recorded data and not from a live customer base.

9.2 Future work

- Use a real telecom dataset, or data from an operator, and repeat the comparison.
- Add time-based features such as the change in usage or bills in the last three months, and try sequence models.
- Tune Random Forest and XGBoost with cross-validation and evaluate over several splits.
- Add cost-sensitive evaluation, in which the cost of an offer and the value of a customer decide the threshold.
- Run a controlled retention experiment (A/B test) to measure the real effect of the actions.
- Monitor the model after deployment for drift and retrain it regularly.

CHAPTER 10
CONCLUSION

This project studied the prediction of telecom customer churn from 13 customer attributes, using a synthetic dataset of 20,000 customers. The problem is important because churn directly reduces revenue and shortens the customer lifetime, and because the operator can act if it knows in advance which customers are at risk.

Logistic Regression, Random Forest and XGBoost were compared on the same stratified split. Logistic Regression obtained the best ROC-AUC (0.8079) with an accuracy of 72.3% and a recall of 72.6%, but the three models were within the uncertainty of the test set, so no model is claimed to be clearly superior. SHAP and the model coefficients agree that contract type, tenure, satisfaction, support calls and complaints are the main drivers. The predicted probability was turned into risk levels that are ordered correctly (9.5%, 33.7% and 64.5% actual churn) and the High-risk group contains 60.5% of the churners.

The next steps are to repeat the study on real data, add history-based features, tune the non-linear models and measure the real effect of the retention actions. Overall, the project shows that good results in AI for engineering systems depend not only on the algorithm but also on understanding the domain, careful data handling, a realistic test, and honest reporting of the errors.

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